

An 80 meter string is hanging from 2 poles that are 50 meters high. The low point of the string is 20 meters above the ground. How far apart are the poles?

We model the hanging string as a **catenary**, which is the correct shape for a uniform flexible cable under gravity.

Step 1 — Set up the catenary

Let the lowest point of the string be at height $y = 0$.

Supports are 30 m higher (since $50 - 20 = 30$).

A catenary has equation

$$y = a(\cosh(x/a) - 1).$$

At the supports ($x = \pm d/2$):

$$a(\cosh(u) - 1) = 30, \quad u = \frac{d}{2a}.$$

Step 2 — Use the total length (80 m)

Length from $-d/2$ to $d/2$:

$$L = 2a \sinh(u) = 80 \quad \Rightarrow \quad a \sinh(u) = 40.$$

Step 3 — Solve for a

We have:

$$\cosh(u) = 1 + \frac{30}{a}, \quad \sinh(u) = \frac{40}{a}.$$

Use the identity $\cosh^2(u) - \sinh^2(u) = 1$:

$$\left(1 + \frac{30}{a}\right)^2 - \left(\frac{40}{a}\right)^2 = 1.$$

Expand:

$$1 + \frac{60}{a} + \frac{900}{a^2} - \frac{1600}{a^2} - 1 = 0,$$

$$\frac{60}{a} - \frac{700}{a^2} = 0,$$

$$60a - 700 = 0 \quad \Rightarrow \quad a = \frac{700}{60} = 11.6667 \text{ m.}$$

Step 4 — Solve for span d

$$\sinh(u) = \frac{40}{a} = \frac{40}{11.6667} = 3.4286.$$

Thus

$$u = \operatorname{asinh}(3.4286) \approx 1.946.$$

Now

$$\frac{d}{2} = au = 11.6667 \times 1.946 \approx 22.7 \text{ m.}$$

So

$$\boxed{d \approx 45.4 \text{ meters}}.$$

Final Answer:

The poles are approximately 45.4 meters apart.

